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United States Patent	12403630
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Ooka; Yusuke et al.

Extruder, dehydration cylinder, dehydrating method and manufacturing method of resin pellet

Abstract

An extruder includes a cylinder, a screw built in the cylinder, a dehydration cylinder block provided in the middle of the cylinder and discharging moisture that is separated from a resin material supplied into the cylinder. The dehydration cylinder block has a structure in which plate-shaped members each having an opening are arranged in a long-axis direction of the cylinder, a screw passing through the opening. Surface roughness of mutually opposing surfaces of the plurality of plate-shaped members is rougher than surface roughness of an inner wall of the cylinder.

Inventors: **Ooka; Yusuke (Tokyo, JP), Uneda; Hiroshi (Tokyo, JP)**

Applicant: **THE JAPAN STEEL WORKS, LTD.** (Tokyo, JP)

Family ID: **83855277**

Assignee: **THE JAPAN STEEL WORKS, LTD. (Tokyo, JP)**

Appl. No.: **17/736058**

Filed: **May 03, 2022**

Prior Publication Data

Document Identifier	Publication Date
US 20220355511 A1	Nov. 10, 2022

Foreign Application Priority Data

JP	2021-078339	May. 06, 2021

Publication Classification

Int. Cl.: **B29B9/16** (20060101); **B29B9/06** (20060101)

U.S. Cl.:

CPC **B29B9/16** (20130101); **B29B9/06** (20130101); B29B9/065 (20130101); B29B2009/168 (20130101)

Field of Classification Search

CPC: B29C (48/761); B29C (48/40); B29C (48/766); B29C (48/0022); B29C (48/68); B29C (48/681); B29C (48/682); B29C (48/767); B29C (48/685); B29C (48/763); B29C (48/29); B29B (9/16); B29B (9/06); B29B (7/489); B29B (7/726); B29B (7/842); B29B (9/065); B29B (13/06); B29B (7/483); B29B (7/90); B29B (2009/168); B29B (2013/002); B29B (2013/005); B29B (7/84); B29B (9/168); B29B (7/482); B28B (3/269); C08C (3/00); Y10S (425/812)

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Primary Examiner: Malekzadeh; Seyed Masoud

Attorney, Agent or Firm: Potomac Law Group, PLLC

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION
(1) The present application claims priority from Japanese Patent Application No. 2021-078339 filed on May 6, 2021, the content of which is hereby incorporated by reference into this application.
TECHNICAL FIELD OF THE INVENTION
(2) The present invention relates to an extruder, a dehydration cylinder, a dehydrating method, and a manufacturing method of a resin pellet.

BACKGROUND OF THE INVENTION

(3) For example, Patent Document 1 (Japanese Patent Application Laid-open No. 2001-129870) discloses a technique about an extruder.

SUMMARY OF THE INVENTION

(4) Resin products such as resin pellets can be manufactured by using a resin material(s) that is extruded from the extruder. When the resin products are manufactured by using the extruder, the resin material is supplied into a cylinder of the extruder, is kneaded and conveyed by a screw built in the cylinder, and is extruded from a die that is attached to a tip portion of the cylinder.

(5) The resin material supplied to the cylinder may contain a high proportion of moisture, but if a water content of the resin material extruded from the extruder is high, it becomes difficult to manufacture resin products such as resin pellets by using the resin material extruded from the extruder. In that case, it is desirable to provide a dehydration cylinder in the middle of the cylinder of the extruder so that moisture separated from the resin material conveyed into the cylinder is discharged from the dehydration cylinder.

(6) However, when the moisture is discharged from the dehydration cylinder, not only the moisture but also a resin component(s) may be discharged from the dehydration cylinder together, which may cause a problem. For this reason, it is desired to selectively discharge the moisture contained in the resin material from the dehydration cylinder and prevent the resin component from being discharged.

(7) Other problems and novel features will become apparent from the description and accompanying drawings herein.

(8) According to one embodiment, an extruder includes a cylinder, a screw built in the cylinder, and a dehydration cylinder portion discharging moisture that is separated from a resin material supplied into the cylinder. The dehydration cylinder portion has a structure in which a plurality of plate-shaped members having opening are arranged in a long-axis direction of the cylinder, the screw passing through each of the opening. Surface roughness of mutually opposing surfaces of the plurality of plate-shaped members is rougher than surface roughness of an inner wall of the cylinder.

(9) According to one embodiment, a dehydrating method includes: (a) supplying a resin material containing moisture into a cylinder; (b) conveying the resin material by a screw in the cylinder; (c) discharging the moisture, which is separated from the resin material, from a dehydration cylinder portion provided in the middle of the cylinder; and (d) extruding the resin material from a die connected at a tip portion of the cylinder. The dehydration cylinder portion has a structure in which a plurality of plate-shaped members each having an opening are arranged in a long-axis direction of the cylinder, the screw passing through the opening. Surface roughness of mutually opposing surfaces of the plurality of plate-shaped members is rougher than surface roughness of an inner wall of the cylinder.

(10) According to one embodiment, the moisture contained in the resin material can be discharged from the dehydration cylinder, and discharging the resin material from the dehydration cylinder can be suppressed or prevented.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is an explanatory diagram showing a configuration example of an extruder according to an embodiment;

(2) FIG. **2** is an explanatory diagram schematically showing a structure inside a cylinder of the extruder of FIG. **1**;

(3) FIG. **3** is a side view showing a dehydration cylinder block of the extruder of one embodiment;

(4) FIG. **4** is a plan view showing a dehydration cylinder block of the extruder of one embodiment;
(5) FIG. **5** is a sectional view showing a dehydration cylinder block of the extruder of one embodiment;
(6) FIG. **6** is a plan view showing a plate-shaped member used in the dehydration cylinder block each shown in FIGS. **3** to **5**;
(7) FIG. **7** is a sectional view of the plate-shaped member shown in FIG. **6**;
(8) FIG. **8** is a sectional view of the plate-shaped member shown in FIG. **6**;
(9) FIG. **9** is a sectional view of the plate-shaped member shown in FIG. **6**;
(10) FIG. **10** is a sectional view showing a plurality of plate-shaped members arranged in a long-axis direction of the cylinder;
(11) FIG. **11** is a partially enlarged sectional view showing a part of FIG. **10** in an enlarged manner;
(12) FIG. **12** is a plan view showing a modification example of a dehydration cylinder block; and
(13) FIG. **13** is a sectional view showing a modification example of a dehydration cylinder block.
DESCRIPTION OF THE PREFERRED EMBODIMENTS
(14) Hereinafter, embodiments will be described in detail with reference to the drawings. Incidentally, through all the drawings for explaining the embodiments, members having the same function are denoted by the same reference numerals, and a repetitive description thereof will be omitted. Further, in the following embodiments, the same or similar parts will not be repeated in principle unless they are particularly necessary.
Embodiment
(15) <Extruder>
(16) FIG. **1** is an explanatory drawing (side view) showing a configuration example of an extruder (extruding machine) **1** of the present embodiment. FIG. **2** is an explanatory diagram (planar perspective view) schematically showing a structure inside a cylinder **2** of the extruder **1**. FIG. **2** shows screws **3** arranged in the cylinder **2** in a perspective manner when the extruder **1** shown in FIG. **1** is viewed from above.
(17) First, a schematic configuration of the extruder **1** will be described with reference to FIG. **1**. The extruder **1** shown in FIG. **1** includes a cylinder (barrel) **2**, two screws **3** rotatably arranged in the cylinder **2**, a rotary drive mechanism **4** for rotating the screws **3** in the cylinder **2**, a hopper (material charging unit, material supplying unit) **5** arranged on an upstream side of the cylinder **2**, and a die (dice, mold) **6** attached to a downstream-side tip of the cylinder **2**. The hopper **5** is connected to an upper surface of the cylinder **2** so that a resin material (raw material, water-containing polymer) **7** can be supplied into the cylinder **2** via the hopper **5**.
(18) Incidentally, when a “downstream side” and an “upstream side” are referred to about the cylinder **2** and the screw **3**, the “downstream side” means a downstream side of a flow of a resin material in the cylinder **2**, and the “upstream side” means an upstream side of the flow of the resin material in the cylinder **2**. Consequently, in the cylinder **2** and the screw **3**, a side closer to a tip of the cylinder **2** is the downstream side, and a side far from the tip of the cylinder **2** is the upstream side. Incidentally, the tip of the cylinder **2** corresponds to an end portion of the cylinder **2** on a side from which the resin material is extruded, that is, an end portion on a side to which the die **6** is connected.
(19) The two screws **3** are rotatably inserted and built in the cylinder **2**. Consequently, the extruder **1** can also be regarded as a twin-screw extruder. In the cylinder **2**, the two screws **3** are arranged so as to mesh with each other and rotate. A long-axis direction of the cylinder **2** and a long-axis direction of the screw **3** in the cylinder **2** are the same and, here, the direction is an X direction. Incidentally, the long-axis direction of the cylinder **2** is a long-side direction or a longitudinal direction of the cylinder **2**, and the cylindrical cylinder **2** extends in the X direction which is the long-axis direction of the cylinder **2**. Further, the long-axis direction of the screw **3** corresponds to an axial direction of a rotation axis of the screw **3**. In the cylinder **2**, the resin material is conveyed from the upstream side to the downstream side in the X direction, which is the long-axis direction,

by the rotating screw **3**.

(20) Further, in each drawing, an X direction, a Y direction and a Z direction are shown as necessary. The X, Y, and Z directions are directions intersecting with each other and, more specifically, are directions orthogonal to each other. Consequently, the X direction and the Y direction are orthogonal to each other, and the Z direction is orthogonal to the X direction and the Y direction. The X direction and the Y direction correspond to a horizontal direction, and the Z direction corresponds to an up-and-down direction (height direction). The X direction is the long-axis direction of the cylinder **2** and, therefore, is also the long-axis direction of the screw **3** in the cylinder **2**.

(21) Further, in the present embodiment, a case where the number of screws **3** in the cylinder **2** is two is described, but, as another embodiment, the number of screws **3** in the cylinder **2** may be one. However, when the number of screws **3** in the cylinder **2** is two, a large space volume can be obtained, so that, in a case of the same screw diameter, where the number of screws **3** is two can enhance supply capacity of the resin material more than where the number of screws **3** is one.

(22) The cylinder **2** is composed of a plurality of cylinder blocks (cylinder portions) **11**, and the plurality of cylinder blocks **11** are arranged and coupled in a direction (here, the X direction) from the upstream side to the downstream side. The cylinder block **11***a*, to which the hopper **5** is connected, among the plurality of cylinder blocks **11** constituting the cylinder **2** has an opening on an upper surface thereof, and the hopper **5** is connected so as to communicate with the opening. Consequently, the resin material **7** charged into the hopper **5** is supplied into the cylinder block **11***a* from the opening on the upper surface of the cylinder block **11***a* to which the hopper **5** is connected.

(23) Further, the plurality of cylinder blocks **11** constituting the cylinder **2** include dehydration cylinder blocks (dehydration cylinders) **11***b*. The dehydration cylinder block **11***b* is provided in the middle of the cylinder **2**. That is, the dehydration cylinder blocks **11***b* are arranged in the middle of the plurality of cylinder blocks **11** that are arranged in a direction from the upstream side to the downstream side. In the cylinder **2**, the dehydration cylinder block **11***b* is arranged on the downstream side of the cylinder block **11***a* to which the hopper **5** is connected. The dehydration cylinder block **11***b* can function as a discharge unit for discharging moisture, which is separated from the resin material **7** supplied into the cylinder **2**, outside the cylinder **2**.

(24) FIG. **1** shows a case where the number of dehydration cylinder blocks **11***b* that the extruder **1** has is two, the cylinder blocks **11***b* being arranged at two locations in the middle of the cylinder **2**. However, the number of dehydration cylinder blocks **11***b* that the extruder **1** has may be changed as needed, and may be any number of 1 or more.

(25) Next, an outline of an operation of the extruder **1** shown in FIG. **1** will be described.

(26) A resin material **7** containing moisture (water) is supplied from the hopper **5** into the cylinder **2**. The resin material **7** supplied to the cylinder **2** contains moisture and a resin component (solid content). The resin material supplied from the hopper **5** into the cylinder **2** is conveyed forward (on the downstream side) in the cylinder **2** by the rotating screw **3**. At this time, the resin material can be kneaded by the rotating screw **3**. While the resin material is being conveyed in the cylinder **2**, moisture is separated from the resin material and the separated moisture is discharged outside from the dehydration cylinder block **11***b*. The resin material (resin material having a reduced water content) conveyed in the cylinder **2** and reaching the tip of the cylinder **2** is discharged from a discharge port of the die **6**. Since the moisture contained in the resin material is discharged outside from the dehydration cylinder block **11***b*, the water content of the resin material discharged from the discharge port of the die **6** is lower than the water content of the resin material **7** at a stage of being supplied from the hopper **5** into the cylinder **2**. A front surface of the die **6**, that is, a surface opposite to a side that is connected to the cylinder **2** is connected to a pelletizer **8**. The resin material discharged from the discharge port of the die **6** is cut one after another by a cutter (cutting blade) **8***a* that the pelletizer **8** has, and is cooled and solidified. Consequently, pellets **9** are formed

as resin pellets. Thereafter, the pellet **9** is conveyed outside the pelletizer **8**, for example, to a dryer. In this way, the pellets **9** can be obtained by using the extruder **1** of the present embodiment. Kneading the pellets **9** with, for example, a functional filler or the like as a secondary raw material makes it possible to obtain a functional pellet with added value. Various resin products are manufactured by using these functional pellets.

(27) As shown in FIG. **2**, the cylinder **2** of the extruder **1** and the screw **3** in the cylinder **2** have the X direction as a long-axis direction. The screw **3** can be configured by combining, as necessary, a screw portion rotating so as to send forth a conveyed object forward at a first speed, a screw portion rotating so as to send forth the conveyed object forward at a speed lower than the first speed, and a screw portion rotating so as to push back the conveyed object backward, and a screw portion arranged so as to prevent the conveyed object from being conveyed forward. Since the screw **3** is configured by combining the various types of screw portions described above, it has such a structure that a part of the cylinder **2** (pressurizing portion **10**) pressurizes the conveyed object.

(28) The dehydration cylinder block **11***b* is arranged on the upstream side of the pressurizing portion **10**. In the resin material conveyed in the cylinder **2**, the moisture in the resin material and the resin component are separated by the pressurizing portion **10**, the resin component is extruded to the downstream side of the pressurizing portion **10**, and the moisture is discharged outside the cylinder **2** from the dehydration cylinder block **11***b*.

(29) In the example shown in FIG. **2**, the pressurizing portion **10** and the dehydration cylinder block **11***b* are provided at two locations in the X direction, respectively. However, most of the moisture contained in the resin material **7** supplied from the hopper **5** into the cylinder **2** shown in FIG. **1** is discharged outside the cylinder **2** from the first dehydration cylinder block **11***b*.

(30) Consequently, a process for manufacturing the pellets **9** by using the extruder **1** includes: a step of supplying the resin material (**7**) containing the moisture into the cylinder **2**; a step of conveying the resin material by the screw **4** in the cylinder **2**; a step of discharging the moisture, which is separated from the resin material, from the dehydration cylinder portion **11***b* provided in the middle of the cylinder **2**; and a step of extruding the resin material from a die **6** connected to the tip portion of the cylinder **2**. The process of manufacturing the pellet **9** by using the extruder **1** further includes a step of cutting the resin material extruded from the die **6** to form the pellet **9**.

(31) <Background of Examination>

(32) The resin material supplied to the cylinder of the extruder may contain a high proportion of moisture, but if the water content of the resin material extruded from the extruder is high, it becomes difficult to manufacture resin products such as resin pellets by using the material extruded from the extruder. For this reason, when the water content of the resin material supplied to the cylinder of the extruder is high, it is desirable to provide a dehydration cylinder (corresponding to the dehydration cylinder block **11***b*) in the middle of the cylinder of the extruder, separate the moisture from the resin material conveyed in the cylinder, and discharge the separated moisture from its dehydration cylinder.

(33) For example, production of polymer such as rubber polymer is generally carried out by emulsion polymerization, solution polymerization, or the like. However, in these final stages, the polymer becomes a slurry(s) containing moisture. Flocculant is charged into the slurry to atomize, as a certain level of lump, fine particles dispersed in the slurry, and aggregated particles and liquid are then separated. The aggregated particles (polymer aggregates, hydrous crumbs) can be used as a resin material (corresponding to the resin material **7**) charged into the hopper of the extruder. In this case, the particles used as the resin material that is charged into the hopper contain moisture and a resin component(s), and the water content becomes high to some extent. As an example, the resin material at a stage of being charged into the hopper contains, for example, about 30 to 50% of moisture. Meanwhile, the water content of the resin material extruded from the extruder is preferably less than 1%, for example. Incidentally, the water content is expressed in percent by

weight.
(34) For this reason, when the dehydration cylinder block is provided in the cylinder of the extruder, it is desired that the dehydration cylinder block can efficiently discharge the moisture contained in the resin material.
(35) However, in dehydrating with the dehydration cylinder block, not only the moisture but also the resin components may be together discharged outside the dehydration cylinder block. If the resin components are also discharged from the dehydration cylinder block, a ratio of the resin components extruded from the extruder among the resin components contained in the resin material supplied to the cylinder of the extruder leads to decreasing. This may bring an increase in manufacturing cost of resin products such as resin pellets. Further, if the resin components are also discharged from the dehydration cylinder block, the resin components may accumulate and clog in a moisture discharge path of the hydration cylinder block. For preventing this, the dehydration cylinder block needs to frequently be cleaned. This brings a reduction in an operating rate of the extruder.
(36) For this reason, when the dehydration cylinder block is provided in the cylinder of the extruder, it is desirable that the moisture contained in the resin material can be selectively discharged in the dehydration cylinder block and, simultaneously, the discharge of the resin components is suppressed.
(37) <Dehydration Cylinder Block>
(38) FIG. **3** is a side view showing a part of the extruder **2** of the present embodiment, FIG. **4** is a plan view (top view) showing a part of the extruder **2** of the present embodiment, and FIG. **5** is a sectional view of the extruder **2** of the present embodiment. FIGS. **3** and **4** show a side view and a plan view of the dehydration cylinder block **11***b*, and a sectional view taken at a position of line A-A shown in FIGS. **3** and **4** substantially corresponds to FIG. **5**. FIG. **6** is a plan view showing a plate-shaped member **13** used in the dehydration cylinder block **11***b*, and a plan view when the plate-shaped member **13** is viewed from the X direction is shown. FIGS. **7** to **9** are sectional views each showing the plate-shaped member **13** used in the dehydration cylinder block **11***b*. A sectional view taken at a position of line B-B shown in FIG. **6** substantially corresponds to FIG. **7**, and a sectional view taken at a position of line C-C shown in FIG. **6** substantially corresponds to FIG. **8**, and a sectional view taken at position of D-D line shown in FIG. **9** substantially corresponds to FIG. **9**. FIG. **10** is a sectional view showing a plurality of plate-shaped members **13** arranged in a long-axis direction of the cylinder **2**, and a sectional view corresponding to a cross-section shown in FIG. **7** is shown. FIG. **11** is a partially enlarged sectional view showing a part of FIG. **10** in an enlarged manner.
(39) In the extruder **1** of the present embodiment, a dehydration cylinder block (dehydration cylinder) **11***b* is provided in the middle of the cylinder **2**, and moisture contained in the resin material **7** supplied from the hopper **5** into the cylinder **2** can be discharged outside the cylinder **2** by the dehydration cylinder block **11***b*.
(40) A structure of the dehydration cylinder block **11***b* will be described with reference to FIGS. **3** to **11**. The dehydration cylinder block **11***b* has a structure in which a plurality of plate-shaped members **13** each having an opening **12** are arranged in the X direction which is the long-axis direction of the cylinder **2**, the screw **3** penetrating through the opening. That is, the dehydration cylinder block **11***b* has the plurality of plate-shaped members **13** arranged in the long-axis direction (X direction) of the cylinder **2**. The plate-shaped member **13** is preferably made of a metal material, for example, stainless steel.
(41) The plate-shaped member **13** has an opening **12** through which the screw **3** passes. Since the plurality of plate-shaped members **13** are arranged in the X direction which is the long-axis direction of the cylinder **2**, the openings **12** of the plurality of plate-shaped members **13** communicate with each other in the X direction. That is, the plurality of plate-shaped members **13** are arranged in the X direction so that the openings **12** communicate with each other. By arranging

the plurality of plate-shaped members **13** each having the opening **12**, a cylindrical cylinder portion (cylinder block **11***b*) is configured.
(42) The screw **3** passes through a space formed by connecting the openings **12** of the plurality of plate-shaped members **13**. For this reason, the plurality of plate-shaped members **13** arranged in the X direction cover an outer periphery of the screw **3**. It is preferable that the respective openings **12** of the plurality of plate-shaped members **13** have the same shape (planar shape) and the same dimensions (planar dimensions). Further, it is preferable that the plurality of plate-shaped members **13** have the same shape and the same dimensions as each other.
(43) In cases of FIGS. **5** and **6**, it is assumed that the number of screws **3** is two, and the opening **12** has such a shape that two circles are partially overlapped. Further, in the cases of FIGS. **5** and **6**, a distance L**1** from an outer periphery of the plate-shaped member **13** to the opening **12** is substantially constant. In other words, a planar shape of the plate-shaped member **13** is such a shape that the distance L**1** from the outer periphery of the plate-shaped member **13** to the opening **12** is substantially constant. This distance L**1** can be set as needed, but can be about, for example, 10 mm. Furthermore, the plate-shaped member **13** is a member whose thickness, a distance between a front surface and a back surface of the plate-shaped member **13**, is thin, and a thickness of the plate-shaped member **13** can be preferably about 0.5 to 5 mm and, for example, can be exemplified as about 1 mm. A thickness direction of the plate-shaped member **13** is the X direction. The thickness of the plate-shaped member **13** described here corresponds to a thickness of a region other than a protrusion **13***a* described later.
(44) A plurality of plate-shaped members **13** arranged in the X direction are fixed to a fixing plate **16** by fixing members **15** such as screws or bolts. Consequently, the plate-shaped member **13** also has an opening **14** for passing through the fixing member **15** such as a screw in addition to the opening **12** for passing through the screw **3**. The fixing plate **16** is a metal member that is thicker in thickness and higher in strength than the plate-shaped member **13**, and has an opening that communicates with the opening **12** of the plate-shaped member **13**. Consequently, the dehydration cylinder block lib has a structure in which the plurality of plate-shaped members **13** arranged in the X direction are sandwiched between a pair of fixing plates **16** separated in the X direction. The fixing plate **16** is preferably made of a metal material, for example, stainless steel.
(45) The screw **3** penetrates, and the resin material conveyed by the rotating screw **3** passes through a space formed by linking the openings **12** of plurality of plate-shaped members **13** in the X direction, that is, a conveyance space of the dehydration cylinder block **11***b*. Hereinafter, the space formed by connecting the openings **12** of the plurality of plate-shaped members **13** in the X direction will be referred to as a conveyance space of the dehydration cylinder block **11***b*. A shape and dimensions of a cross-section (cross-section perpendicular to the X direction) of the conveyance space of the dehydration cylinder block **11***b* substantially correspond to the planar shape and dimensions of the opening **12**. Inner walls of the openings **12** of the plurality of plate-shaped members **13** arranged in the X direction constitute an inner wall of the conveyance space of the dehydration cylinder block **11***b*. Further, in the cylinder block **11** other than the dehydration cylinder block **11***b*, a space through which the screw **3** penetrates and the resin material conveyed by the rotating screw **3** passes will be referred to as a conveyance space of the cylinder block **11**. The conveyance space of the cylinder block **11** on the upstream side of the dehydration cylinder block **11***b*, the conveyance space of the dehydration cylinder block **11***b*, and the conveyance space of the cylinder block **11** on the upstream and downstream sides of the dehydration cylinder block **11***b* communicate with one another in the X direction, and the shapes and dimensions of their cross-sections (cross-sections perpendicular to the X direction) can be substantially the same as one another.
(46) The resin material supplied from the hopper **5** into the cylinder **2** is conveyed on the downstream side in the cylinder **2** by the rotation of the screw **3**. At this time, the resin material passes from the conveyance space of the cylinder block **11** on the upstream side of the dehydration

cylinder block **11***b* through the conveyance space of the dehydration cylinder block **11***b* and is sent to the conveyance space of the cylinder block on the downstream side of the dehydration cylinder block **11***b*.
(47) In the dehydration cylinder block **11***b*, the moisture separated from the resin material can be discharged outside from between mutually opposing surfaces of the plurality of plate-shaped members **13**. That is, the moisture separated from the resin material conveyed in the cylinder **2** by the rotating screw **3** can be discharged outside through a gap **17** between the mutually opposing surfaces of the plurality of plate-shaped members **13**. The gap **17** can function as a slit for discharging the moisture separated from the resin material, and serves as a flow path (discharge path) for the moisture separated from the resin material.
(48) Here, the moisture separated from the resin material can be discharged through the gap **17** between the plurality of plate-shaped members **13**, but the resin component(s) contained in the resin material is desirably prevented from being discharged from the gap **17** between the plurality of plate-shaped members **13** as much as possible. That is, in the dehydration cylinder block **11***b*, it is desirable that the moisture separated from the resin material is selectively discharged from between the mutually opposing surfaces of the plurality of plate-shaped members **13**.
(49) The moisture is discharged from the gap **17**, but it is effective to increase static pressure of the gap **17** (resistance when a fluid passes through the gap **17**) in order to prevent the resin component from being discharged. This is because when the static pressure of the gap **17** is large, in comparison with the moisture and the resin component, the resin component having relatively high viscosity does not invade (penetrate into) the gap **17** and the moisture having low viscosity selectively invades the gap **17**.
(50) Thus, in the present embodiment, surface roughness of the mutually opposing surfaces (surfaces opposing the X direction) of the plurality of plate-shaped members **13** arranged in the X direction is roughened (see FIG. **11**). That is, the mutually opposing surfaces of the plurality of plate-shaped members **13** are subjected to a roughening treatment. This makes it possible to increase the static pressure of the gap **17**.
(51) If the surface roughness of the mutually opposing surfaces of the plurality of plate-shaped members **13** is low, a distance (interval) between the mutually opposing surfaces of the plurality of plate-shaped members **13** is almost the same (constant) regardless of positions in the surface. Meanwhile, when the surface roughness of the mutually opposing surfaces of the plurality of plate-shaped members **13** is roughened (large), a large number of minute irregularities are present on the roughened surface and, by reflecting such a situation, the distance between the mutually opposing surfaces of the plurality of plate-shaped member **13** varies depending on the positions in the surface (see FIG. **11**).
(52) Since the surface roughness of the mutually opposing surfaces of the plurality of plate-shaped members **13** is rough (roughened), the gap **17** between the mutually opposing surfaces of the plurality of plate-shaped members **13** is configured so that a cross-sectional area as a flow path of the moisture changes in a complicated manner. Consequently, when the mutually opposing surfaces of the plurality of plate-shaped members **13** are roughened (when the surface roughness is rough), as compared with a case where the surface roughness is not roughened (when the surface roughness is low), the static pressure of the gap **17** which becomes the discharge path of the moisture can be increased. As described above, when the static pressure of the gap **17** is large, in comparison with the moisture and the resin component the resin component having relatively high viscosity does not invade the gap **17** and the moisture having low viscosity selectively invade the gap **17**. As a result, the moisture separated from the resin material can be discharged outside from the gap **17**, and the resin component contained in the resin material can avoid leaking outside through the gap **17**.
(53) The surface roughness of the mutually opposing surfaces of the plurality of plate-shaped members **13** preferably has 1.6a to 25a (1.6a or more and 25a or less) in terms of arithmetic mean roughness Ra. The arithmetic mean roughness Ra can be measured as follows. First, irregularities

(unevenness) of a surface in a measurement section (length) to be targeted is measured. Next, an average value of the measured irregularities is set as a reference line, and a difference between the reference line and an irregularity curve is integrated along the measurement section. A value obtained by dividing this integration result by a length of the measurement section is the arithmetic mean roughness Ra.

(54) Further, the mutually opposing surfaces of the plurality of plate-shaped members **13** constituting the dehydration cylinder block **11***b* is subjected to the roughening treatment, but the inner wall (inner surface) of the cylinder **2** (cylinder block **11**) is subjected to no roughening treatment and surface roughness of the inner wall of the cylinder **2** is lower than the surface roughness of the mutually opposing surfaces of the plurality of plate-shaped members **13**. This is because the inner wall of the cylinder **2** constitutes the inner wall of the space in which the resin material is conveyed by the rotating screw **3**, so that the mutually opposing surfaces, which is subjected to no roughening treatment, are more suitable so as not to adversely affect the conveyance of the resin material. Consequently, in the present embodiment, the surface roughness of the mutually opposing surfaces of the plurality of plate-shaped members **13** constituting the dehydration cylinder block **11***b* becomes rougher (larger) than the surface roughness of the inner wall of the cylinder **2**. Here, the surface roughness of the inner wall of the cylinder **2** corresponds to the surface roughness of the inner wall of the cylinder **2** (cylinder block **11**) other than the dehydration cylinder block **11***b*.

(55) Further, in the plurality of plate-shaped members **13** constituting the dehydration cylinder block **11***b*, the surface roughness of at least one of both surfaces (two surfaces located on opposite sides to each other) of each plate-shaped member **13** is preferably roughened (large) (i.e., subjected to the roughening treatment), and it is more preferable that the surface roughness of the both surfaces is roughened (subjected to the roughening treatment). Consequently, the surface roughness of at least one of the two surfaces forming the gap **17**, more preferably, the surface roughness of the both surfaces becomes roughened, so that the static pressure of the gap **17** is increased to selectively remove the moisture from the gap **17** so that the resin component leaking outside through the gap **17** can be suppressed and prevented.

(56) In addition, in the present embodiment, the gap **17** between the plurality of plate-shaped members **13** arranged in the X direction serves as a flow path of the moisture separated from the resin material. Consequently, a flow-path length when the moisture passes through the gap **17** can be easily adjusted by adjusting the shape and dimensions of the plate-shaped member **13**. For example, in the case of FIG. **6**, a distance L**1** from the outer circumference of the plate-shaped member **13** to the opening **12** becomes a flow-path length when the moisture passes through the gap **17**. Consequently, by adjusting the distance L**1**, controlled can be discharge efficiency when the moisture separated from the resin material is discharged outside through the gap **17**. For example, if the distance L**1** is too large, the discharge efficiency of the moisture through the gap **17** may decrease. However, by reducing the distance L**1** to some extent, for example, by setting the distance to about 20 mm or less, it becomes easier to secure the discharge efficiency of the moisture through the gap **17**.

(57) Further, since each of the plurality of plate-shaped members **13** is thin in thickness, each mechanical strength thereof is not so high. However, by arranging the plurality of plate-shaped members **13** in the X direction, the overall mechanical strength of the plurality of plate-shaped members **13** can be enhanced.

(58) Furthermore, in the cases of FIGS. **6** to **9**, each of the plurality of plate-shaped members **13** has a protrusion **13***a* that locally protrudes in the long-axis direction (X direction) of the cylinder. The protrusion **13***a* can also be regarded, in the plate-shaped member **13**, as a portion thinner in thickness than a region other than the protrusion **13***a*. In the cases of FIGS. **6** to **9**, a case where the opening **14** through which the fixing member **15** passes is formed in the protrusion **13***a* is shown and, in the plate-shaped member **13**, a peripheral region of the opening **14** is thicker (larger) in

thickness than other regions. The protrusion **13***a* may be integrally formed with the plate-shaped member **13**, or may be formed by joining another member, for example, a film member such as a metal leaf, to the plate-shaped member **13** having a uniform thickness. Moreover, the protrusion **13***a* can be formed by punching or the like.
(59) The protrusion **13***a* can define the gap **17** between the plurality of plate-shaped members **13**. That is, the dimension (dimension in the X direction) of the gap **17** between the plurality of plate-shaped members **13** can be defined by a protrusion amount H**1** of the protrusion **13***a*. When the protrusion amount H**1** of the protrusion **13***a* is small, the dimension of the gap **17** in the X direction becomes small, and when the protrusion amount H**1** of the protrusion **13***a* is large, the dimension of the gap **17** in the X direction becomes large. The larger the dimension of the gap **17** in the X direction becomes, the higher the discharge efficiency of the moisture through the gap **17** becomes. However, if the dimension of the gap **17** in X direction increases, the resin component may leak outside through the gap **17**. Consequently, it is desirable that the dimension of the gap **17** in the X direction is set to an appropriate dimension according to characteristics etc. of a raw material **7** supplied to the cylinder **2**.
(60) In the present embodiment, since the gap **17** between the plurality of plate-shaped members **13** can be defined by the protrusion **13***a*, setting the protrusion amount H**1** of the protrusion **13***a* according to kinds and characteristics, etc. of the resin material supplied to the cylinder **2** makes it possible to adjust the dimension of the gap **17** in the X direction to the optimum dimension. This makes it possible to balance: enhancement of the discharge efficiency when the moisture separated from the resin material is discharged outside from the gap **17**; and prevention of the resin component contained in the resin material from leaking outside through the gap **17**. The protrusion amount H**1** of the protrusion **13***a* can be, for example, about 0.01 to 1 mm.
(61) Further, even if internal pressure of the cylinder **2** changes, the dimension of the gap **17** between the plurality of plate-shaped members **13** in the X direction hardly changes. Consequently, only the moisture separated from the resin material can be stably discharged from the gap **17** between the plurality of plate-shaped members **13**.
(62) Furthermore, the present embodiment may have a configuration of not providing the protrusion **13***a*, that is, of setting the protrusion amount H**1** of the protrusion **13***a* to zero. Even when the protrusion **13***a* is not provided, the surface roughness of the mutually opposing surfaces of the plurality of plate-shaped members **13** is roughened, so that a large number of minute irregularities are present on the surfaces of the plate-shaped members **13**, which makes it possible to secure the flow path of the moisture between the mutually opposing surfaces of the plurality of plate-shaped members **13**. Thus, even if the protrusion **13***a* is not provided, the moisture separated from the resin material can be discharged outside through the gap **17**.
(63) FIGS. **12** and **13** are a plan view (FIG. **12**) and a sectional view (FIG. **13**) each showing a modification example of the dehydration cylinder block **11***b*. FIG. **12** is a bottom view, FIG. **13** shows a cross-section corresponding to FIG. **5**, and a section view taken at a position of line A**1**-A**1** shown in FIG. **12** substantially corresponds to FIG. **13**.
(64) In cases of FIGS. **12** and **13**, the plurality of plate-shaped members **13** arranged in the X direction are covered with a metal cover member (metal member, metal block) **21**. Covering the plurality of plate-shaped members **13** arranged in the X direction with the metal cover member **21** makes it possible to protect the plate-shaped members **13**. The cover member **21** covers the outer periphery of the plurality of plate-shaped members **13** arranged in the X direction, but has an opening **22** for discharging moisture. Consequently, the moisture discharged from the gap **17** between the plurality of plate-shaped members **13** can be discharged outside through the opening **22** of the cover member **21**.
(65) Although the invention made by the present inventor(s) has been specifically described above based on the embodiments thereof, the present invention is not limited to the above-mentioned

embodiments and, needless to say, can be variously modified without departing from the scope thereof.

Claims

1. An extruder comprising: a cylinder; a supplying unit supplying a resin material containing moisture into the cylinder; a screw built in the cylinder and conveying the resin material supplied to the cylinder; and a dehydration cylinder portion provided in middle of the cylinder and discharging the moisture separated from the resin material, wherein the dehydration cylinder portion has a structure in which a plurality of plate-shaped members, each having an opening, are arranged in a long-axis direction of the cylinder, the screw passing through the opening; an outer periphery of the plurality of plate-shaped members is covered by a cover member, the cover member including a plurality of openings for discharging the moisture from the dehydration cylinder portion; surface roughness of mutually opposing surfaces of the plurality of plate-shaped members is rougher than surface roughness of an inner wall of the cylinder; the surface roughness of the mutually opposing surfaces of the plurality of plate-shaped members is 1.6a to 25a in terms of arithmetic mean roughness Ra; a distance between the outer periphery of each of the plate-shaped members and corresponding openings in the plate-shaped members is 10 mm to 20 mm; and a distance between a front surface and a back surface of each plate-shaped member along the long-axis direction is 0.5 mm to 5 mm.
2. The extruder according to claim 1, wherein the moisture separated from the resin material is discharged from between the mutually opposing surfaces of the plurality of plate-shaped members.
3. The extruder according to claim 1, wherein each of the plurality of plate-shaped members has a protrusion that protrudes in the long-axis direction of the cylinder, and a gap between the plurality of plate-shaped members is defined by the protrusion.
4. The extruder according to claim 1, comprising two screws built in the cylinder so as to mesh with each other and so as to convey the resin material supplied to the cylinder, the opening in each of the plate-shaped members having a shape of two partially overlapping circles through which corresponding screws pass.